The impact of deep slow slip on megathrust seismic cycles

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Key Points:

- Simulations show that deep slow slip erodes the long-term seismic moment budget of the updip megathrust through transient creep.
- Reduced long-term seismic moment updip of slow slip is balanced by increased seismic release in adjacent megathrust segments.
- The reorganized seismic moment budget appears geodetically as a coupling reduction of 0.25-0.50, even as the megathrust continues to host $M_w \geq 7$ earthquakes

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Abstract

Geodetically-inferred plate boundary slip-deficit rates constrain seismic hazard by mapping where megathrust faults are coupled and accumulating stress ahead of future ruptures. How transient loading from deep slow slip affects the recovered plate coupling and the seismic behavior of the megathrust remains unclear. Using rate- and state-dependent seismic cycle simulations of the Guerrero segment of the Mexican subduction zone, we find that deep slow slip can simultaneously reduce seismic moment release in the updip seismogenic zone and increase it in adjacent along-strike megathrust segments. This mechanical interaction produces a reduction in the recovered geodetic coupling of 0.25 to 0.50, similar to observations of the poorly coupled Guerrero margin. Despite this low coupling, the megathrust remains capable of hosting $M_w \geq 7$ earthquakes consistent with historical seismicity. Our results suggest low geodetic coupling near recurring slow slip signals persistent, rather than reduced, seismic hazard.

Plain Language Summary

The largest earthquakes happen on the giant megathrust faults where one tectonic plate dives beneath another. Between earthquakes, parts of these faults remain stuck while plate motion continues, slowly building the stress that will later be released in an earthquake. Scientists use precise satellite measurements of ground motion to map which parts of a fault are stuck, or "coupled," because stuck patches are thought to building stress before a future earthquake. But faults can also host slow slip, where a deep part of the fault quietly slides over weeks to months without shaking the ground. Faults near regions of slow slip often look only weakly stuck at the surface, which is often interpreted to suggest a reduced risk of earthquakes. Yet past large earthquakes in Mexico, New Zealand, Alaska, and Japan ruptured such weakly stuck segments. Using computer simulations of the Guerrero region in Mexico, we show that deep slow slip does not switch earthquakes off; it mainly moves them around, reducing earthquakes directly above where slow slip happens while increasing them on neighboring segments. Our results suggest that recurring slow slip can create the appearance of a poorly coupled fault that remains capable of producing major earthquakes.

1 Introduction

Two distinct modes of fault slip, slow slip and earthquakes accommodate relative plate motion at different depths along plate boundary faults (Peng & Gomberg, 2010; Dragert et al., 2001). An outstanding question is how these different but neighboring faulting regimes mechanically interact. Slow slip can advance the timing of nearby earthquakes by transferring stress onto adjacent fault segments (Radiguet et al., 2016; Socquet et al., 2017; Cruz-Atienza et al., 2021), while earthquakes can in turn trigger or modulate slow slip at distances of tens to hundreds of kilometers (Zigone et al., 2012; Wallace et al., 2017; Maubant et al., 2020, 2025). The largest slow slip events regularly occur downdip and adjacent to the seismogenic megathrust (Takagi et al., 2019; Rousset, 2019), such as those in the Mexican subduction zone, the Hikurangi margin (New Zealand), beneath the Bungo Channel (Japan), and in the Atacama region (Chile) (Radiguet et al., 2012; Wallace & Beavan, 2010; Klein et al., 2018, 2023; Takagi et al., 2019). Our work here explores the consequences of this spatial proximity to ask the question, how does slow slip adjacent to a seismogenic source region impact its long-term seismic behavior?

Quantifying this interaction is observationally challenging, as the geodetic record spans at most several decades while seismic cycles operate over centuries to millennia. We can obtain a geological "snapshot" of the kinematic coupling of the megathrust with modern geodetic observatories (Avouac, 2015; Lindsey et al., 2021; Maubant et al., 2023). Plate coupling ϕ is defined as the deficit of the fault slip rate relative to the long-term rate of

relative motion,

$$\phi = 1 - \frac{V}{V_{pl}}, \tag{1}$$

where V is the fault slip rate and V_{pl} the long-term plate convergence rate, so that $\phi = 1$ corresponds to zero interseismic slip and $\phi = 0$ to slip at the plate rate. Plate coupling is a kinematic measure to identify regions accumulating slip deficit, but does not necessarily imply a frictionally unstable interface (K. Wang & Dixon, 2004; Sato et al., 2025). This simple framework of plate coupling implicitly treats V as a single, time-invariant slip rate over the geodetic observation window, with no degree of freedom to represent transient slip. Slow slip is instead accounted for by a slower constant V and recovered as an apparent reduction in coupling, even when the fault remains fully coupled between transients (Mouchon et al., 2023; Maubant et al., 2023).

In subduction zones that host deep slow slip, geodetic inversions reveal a persistent pattern of a shallow seismogenic zone updip of slow slip that appears less coupled than neighboring segments at similar depths (Radiguet et al., 2012; Wallace & Beavan, 2010; Takagi et al., 2019). Such regions of low coupling are often considered to pose reduced seismic hazard (Avouac, 2015; Kaneko et al., 2010). This interpretation rests on an implicitly zero-sum view of the slip budget. At each point on the interface, plate motion is partitioned between an accumulating slip deficit that will eventually be recovered through earthquakes and creep that accommodates plate motion aseismically, so that coupling < 1 implies less stored elastic strain, and a reduced seismic moment available for future ruptures. This accounting only holds if the seismic moment of a weakly coupled segment is genuinely reduced compared to what would be expected for a fully coupled fault. The historical and instrumental earthquake records contradict this model of stationary plate coupling. In the Hikurangi margin, the 1947 Gisborne earthquakes (Mw 7.0–7.1) ruptured a segment that is now considered weakly coupled and regularly hosts slow slip events every 2–3 years (Bell et al., 2014). In the Mexican subduction zone, the Guerrero seismic gap likely hosted a Mw 7.5–8 earthquake in 1911 despite observations of persistent low coupling (Singh et al., 1981; Radiguet et al., 2016; Maubant et al., 2022). The 2021 Mw 8.2 Chignik earthquake in Alaska ruptured a segment of relatively low coupling where shallow slow slip had been detected three years prior (He et al., 2023). Temporal variability of geodetically-inferred coupling has also been documented in multiple regions (Nishimura et al., 2004; Materna et al., 2019; Maubant et al., 2023), notably in Mexico where Villafuerte et al. (2025) related changes in megathrust coupling due to slow slip activity to the earthquake potential of the Oaxaca segment. These observations suggest that the geodetic coupling of seismogenic segments near slow slip may not directly reflect the long-term seismic potential of the plate boundary.

Here, we use seismic cycle simulations governed by slip-rate- and state-dependent friction (Barbot, 2019) to investigate how deep slow slip modifies the seismic cycle of the updip seismogenic megathrust. We use the Guerrero segment of the Mexican subduction zone as a natural laboratory due to its combination of recurring slow slip (Radiguet et al., 2012; Cruz-Atienza, Franco, et al., 2025), low geodetic coupling (Radiguet et al., 2016; Maubant et al., 2022), and historical evidence for large earthquakes in a segment that has not ruptured in over a century (Radiguet et al., 2012; Maubant et al., 2022; Cruz-Atienza, Franco, et al., 2025; Singh et al., 1981; Husker et al., 2023). Previous rate-and-state models have already addressed the Guerrero interface at the scale of individual cycles. Perez-Silva et al. (2021) developed a three-dimensional model of the long-term slow slip cycle, and Li and Gabriel (2024) linked the slow slip cycle to the nucleation of the 2014 M_w 7.3 Papanoa earthquake. We build on this work by focusing on how slow slip affects plate boundary earthquake behavior over many cycles. We compare two simulations spanning more than a thousand years of large megathrust earthquakes with and without slow slip downdip of the seismogenic zone. These long-term simulations allow us to explore the behavior of the megathrust beyond the time scale of the geodetic record. We observe models incorporating slow slip exhibit low coupling in the seismogenic segment directly updip of the slow slip zone

throughout the interseismic period of the seismic cycle, rather than confined to the years immediately preceding rupture. These results demonstrate how seismic hazard can be sustained in regions of low coupling, reconciling current geodetic observations of low coupling and historical evidence of megathrust earthquakes in several major subduction zones.

2 Models and Method

To isolate the mechanical effect of recurring slow slip on the seismic cycle, we develop two simplified simulations of the Guerrero megathrust. The geometry of the shallow seismogenic region is guided by historical ruptures, while the location and properties of the deep slow slip source are calibrated against the first-order characteristics of recurring Guerrero slow slip events. We refer to the simulation including this prescribed slow slip source as the EQ+SS model. We then construct an otherwise identical simulation without the prescribed slow slip source, referred to as the "EQ-only" model; this denotes the absence of the prescribed deep source of Guerrero-like slow slip, but not the absence of all aseismic slip transients.

We use the quasi-dynamic boundary-integral method with radiation damping implemented in the Unicycle software (Barbot, 2019) developed to model plate boundary systems (L. Wang & Barbot, 2020; B. Wang & Barbot, 2024; Guvercin & Barbot, 2025; M. Liu et al., 2026) and validated in community benchmarks (Erickson et al., 2020; Jiang et al., 2022). The fault is a simplified piecewise-planar approximation of the Guerrero megathrust ((Figure 1) with a shallowly dipping Cocos plate that steepens landward and subsequently flattens near 40 km depth (Pérez-Campos et al., 2008; Radiguet et al., 2012); the fault is discretized into 220×104 elements of 2×2 km, Text S1). We run each simulation for 1,600 years and discard the first 500 years to avoid the influence of the initial conditions. Fault dynamics is controlled by slip-rate and state-dependent friction (Barbot, 2019). The fault is loaded through a backslip formulation, in which the traction rate on each element is derived from the elastic interaction of the slip-rate deficit relative to a fixed long-term convergence rate (Barbot, 2019; M. Liu et al., 2026). Plate motion therefore loads every element rather than being transmitted through the surrounding rate-strengthening region, and an element slipping steadily at V_{pl} experiences no net stressing (Text S1).

The constitutive parameters a and b control the slip-rate and state dependence of the friction. Regions with $a > b$ are rate-strengthening and respond stably to slip rate perturbations, whereas regions with $a < b$ are rate-weakening and can host transient slip. $a - b$ controls the local frictional response, whereas coupling is a kinematic measure of the slip rate relative to plate convergence. We do not infer the sign of $a - b$ from geodetic coupling, because a rate-strengthening region can remain highly coupled through its mechanical interaction with the surrounding fault. Both earthquake and slow slip rate-weakening regions have $a = 0.01$ and $b = 0.014$ ($a/b = 0.71$) and $G = 30$ GPa. Their dimensions are relative to the aging-law nucleation length h^* of Rubin and Ampuero (2005), appropriate for $a/b > 0.5$ rather than the classical linear-stability length of Rice and Ruina (1983), and thus control their stability regime without uniquely determining their slip behavior (Texts S1–S3; Figures S1–S3; Table S2).

A shallow, elongated seismogenic zone extends about 400 km along strike and 44 km downdip along the fault surface ($16.0h^* \times 1.8h^*$) between the Michoacán and southern Guerrero historical rupture areas (Figure 1b). Because the spatial regularization of geodetic coupling models smooth estimates of slip deficit, we do not use geodetic models to define frictional boundaries. The downdip width of the seismogenic region is instead guided by the extent of historical earthquake ruptures (Kostoglodov & Pacheco, 1999). The shallow seismogenic zone has uniform frictional properties along its full along-strike extent. The entire fault is driven by a long-term plate convergence rate of 6.1 cm/yr (DeMets et al., 2010).

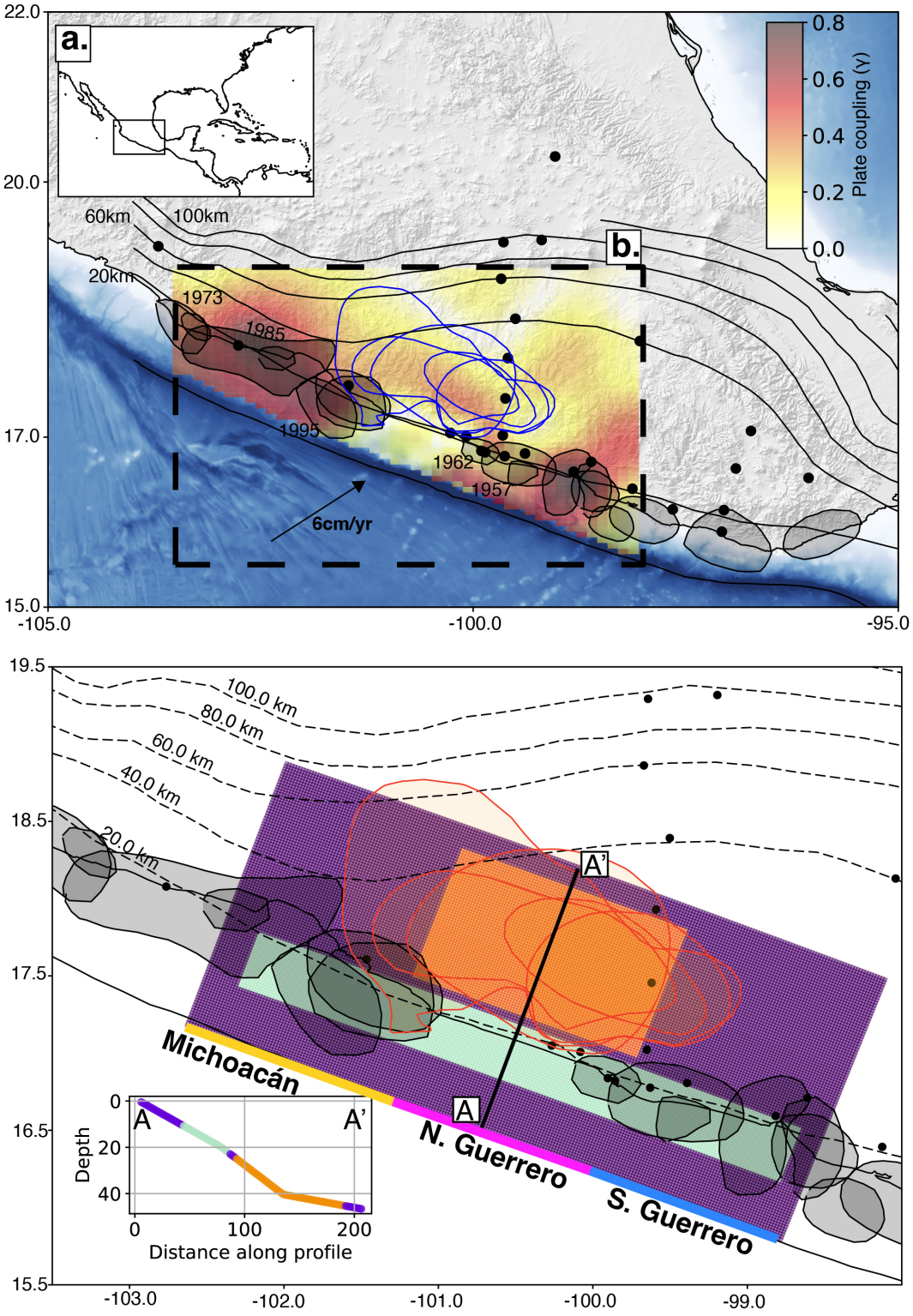


Figure 1. Model fault geometry within the context of the tectonic setting of the Mexican subduction zone and its historical seismic activity. a) Place coupling across the Mexican subduction zone. Colors show the geodetic coupling coefficient inferred by Radiguet et al. (2016) from the GNSS network shown by black dots. Blue contours indicate the spatial extent of previous slow slip events (Radiguet et al., 2012, 2016; Maubant et al., 2020), and black contours show historical earthquake rupture areas (Kostoglodov & Pacheco, 1999). b) Zoom on the Guerrero region showing the fault geometry and frictional domains used in our simulations. The purple region is a uniform rate-strengthening background ($a - b = 0.004$). The shallow green region is a rate-weakening seismogenic domain whose frictional properties are uniform along strike ($a - b = -0.004$). The orange region is the localized rate-weakening source used to generate recurring Guerrero slow slip events ($a-b = -0.004$). Their detailed boundaries and frictional properties are not mapped directly from the coupling model (a). c) Fault-plane profile along A–A′ showing the same frictional domains as a function of depth.

2.1 EQ-only model

In the EQ-only control model, the prescribed deep zone of slow slip is absent and the deeper interface is uniformly rate-strengthening. We observe that a persistent along-strike segmentation of the updip seismogenic zone emerges in the simulated earthquake sequence that broadly corresponds to the Michoacán, northern Guerrero, and southern Guerrero segments (Figures 1 and S5). These three along-strike regions are not prescribed through along-strike frictional boundaries within the shallow seismogenic zone. A homogeneous fault whose dimension is large relative to the characteristic nucleation scale spontaneously exhibits such segmentation (Cattania, 2019). We therefore use these three regions as diagnostic windows on the spatial organization that emerges in the EQ-only model without needing to prescribe any frictional boundaries.

2.2 EQ+SS model

A prescribed slow slip source is introduced in the EQ+SS model downdip of the seismogenic zone at approximately 24–45 km depth. The source is tuned to reproduce the first-order characteristics of the Guerrero slow slip events that occur about every 4 years (Text S2; Figures S2–S3). The earthquake and slow slip rate-weakening regions are separated by a rate-strengthening buffer approximately 6 km wide along the fault surface with the same frictional properties as the surrounding stable fault (Text S1). The location of this source is also consistent with the observed along-strike organization of slow slip in southern Mexico, where Guerrero and Oaxaca host distinct regions of recurring slow slip activity separated by a region of reduces slow slip (Graham et al., 2016; El Yousfi et al., 2023; Cruz-Atienza, Franco, et al., 2025). We focus on the principal recurring Guerrero slow slip source to isolate its interaction with the overlying seismogenic zone.

We calibrate the two unstable domains (seismogenic and slow slip source regions) separately before combining them into the EQ+SS model (Texts S1–S3; Figures S1–S4), while ensuring the seismogenic domain is identical in both models. The parameters of the slow slip source region, $L = 8$ mm and $\bar{\sigma} = 2.7$ MPa ($h^* = 66.0$ km) reproduce the first-order recurrence, duration, magnitude, and slip of Guerrero slow slip events (Radiguet et al., 2012). The low effective normal stress is an effective parameterization of the slip source rather than an inference of the along-strike distribution of pore-fluid pressure (Y. Liu & Rice, 2007). In the seismogenic zone, $L = 9$ cm and $\bar{\sigma} = 80$ MPa ($h^* = 25.1$ km) produce earthquake magnitudes of about M_w7 and recurrence intervals shorter than 100 years (Singh et al., 1981; Husker et al., 2023). All model parameters are listed in Table S2.

3 Results

We observe that adding slow slip modifies the spatial and temporal organization of major earthquakes. Over the simulated 1600 years, the two models produce comparable numbers of $M_w \geq 7$ ruptures with similar magnitudes (M_w 7.0–7.7), consistent with the historical Mexican catalog (Singh et al., 1981; Husker et al., 2023), but differ in the spatial distribution of cumulative seismic moment release (Figure 2a,b; Figure S5).

Introducing the localized slow slip source beneath northern Guerrero modifies the partitioning of rupture activity and cumulative seismic moment release among the three regions. The most visible effect of slow slip is a redistribution of seismic moment release along strike (Figure 2b). In the EQ-only model, seismic release is concentrated in the central segment of northern Guerrero, where two strong peaks near 190 and 260 km along strike correspond to repeated ruptures. When the deep slow slip source is included, these central peaks are strongly reduced as seismic moment is pushed towards the neighboring Michoacán and southern Guerrero segments. A two-dimensional map of the cumulative seismic moment difference confirms that the reduction is concentrated in northern Guerrero, whereas the increases in the neighboring regions are spatially localized (Figure S10). A complementary

reorganization is visible in the cumulative aseismic moment (Figure 2a). In the EQ-only model, troughs in aseismic moment in central northern Guerrero coincide with the peaks of seismic release, consistent with stationary plate coupling (Eq. 1): where the fault ruptures repeatedly, a smaller fraction of the slip budget is accommodated aseismically. When slow slip is introduced downdip in the EQ+SS model, the central segment releases more moment aseismically than in the EQ-only model.

To quantify this reorganization, we integrate the moment released between 500 and 1500 years in each of the three analysis regions (Table S1). Slip transients with $V \geq 10^{-3}$ m/s are counted as seismic, while anything slower is considered aseismic. Slow slip events are identified separately as aseismic transients with onset and termination slip rates of 10^{-8} m/s ($\approx 5\,V_{pl}$; Text S2). We find that the aseismic budget dominates in every region. Seismic slip accounts for 10.8% of the total moment in northern Guerrero and only 1.8% and 1.3% in Michoacán and southern Guerrero (Table S1). To quantify how the moment partitioning between seismic and aseismic slip in the two models, we can compare the moment ratios $R = M_0^{EQ+SS}/M_0^{EQ}$ of the two simulations, computed separately for seismic (R_{seis}), aseismic (R_{aseis}), and total (R_{tot}) moment, where $R = 1$ indicates no difference (Figure 2c). We find that slow slip reduces the seismic moment updip in northern Guerrero with $R_{seis} = 0.41$, while the Michoacán and southern Guerrero release approximately twice as much cumulative seismic moment once slow slip is introduced ($R_{seis} = 2.05$ and 2.40; Figure S10). These ratios are stable across time windows (Figure S7), confirming a systematic mechanical effect rather than a result of stochastic variability. Because the aseismic budget is at least an order of magnitude larger than the seismic budget, a small change in the aseismic moment ($R_{aseis} = 0.97$–1.04) can offset a major ∼50% change in the seismic moment budget. Despite these changes, the total moment is nearly conserved in each region ($R_{tot} = 0.97$–0.99; Figure S6). This is not surprising as the total moment budget is prescribed by the long-term slip rate that is independent of fault behavior. We note that the increase of seismic moment in the neighboring segments is not entirely offset by the decrease in the central segment, with an increase of the aseismic budget in the central segment that takes up the remainder. We conclude that deep slow slip primarily acts to redistribute the seismic moment partitioning in the overlying seismogenic zone, even as large earthquakes continue to occur.

4 Slow slip and megathrust plate coupling

4.1 In situ plate coupling

While slow slip does not prevent major earthquakes, it modifies stress accumulation and release within the seismogenic zone during the interseismic period with two distinct expressions: a reduction of the time-averaged coupling and temporal coupling variations at the slow slip recurrence period. We can quantify both via the in situ coupling of the seismogenic megathrust (without the deep slow slip zone) using the time-resolved form of Eq. 1,

$$\phi(t) = 1 - \frac{u(t + \Delta t) - u(t)}{V_{pl}\,\Delta t}, \tag{2}$$

where u is the slip accumulated on a given seismogenic patch over Δt (Text S4). We use "coupling" for this kinematic slip-deficit quantity that we do not interpret as a measure of frictional stability (K. Wang & Dixon, 2004; Sato et al., 2025). Figure 3 presents this single quantity in two complementary ways. Panels (a–b) show its temporal evolution that we spatially averaged over the rupture area of each earthquake (estimated as the area that slips > 20 cm within the 15–22 km seismogenic depth range; Texts S3, S4) and stacked relative to earthquake occurrence; $t = 0$ is defined as the earthquake time (Text S4). Panels (c–d) map out this coupling across the fault as the median of every 40–10 yr pre-earthquake time window. The 30-year span of the 40–10 yr interval is comparable in duration to the modern geodetic record in Mexico that has captured more than 5 cycles of slow slip. By

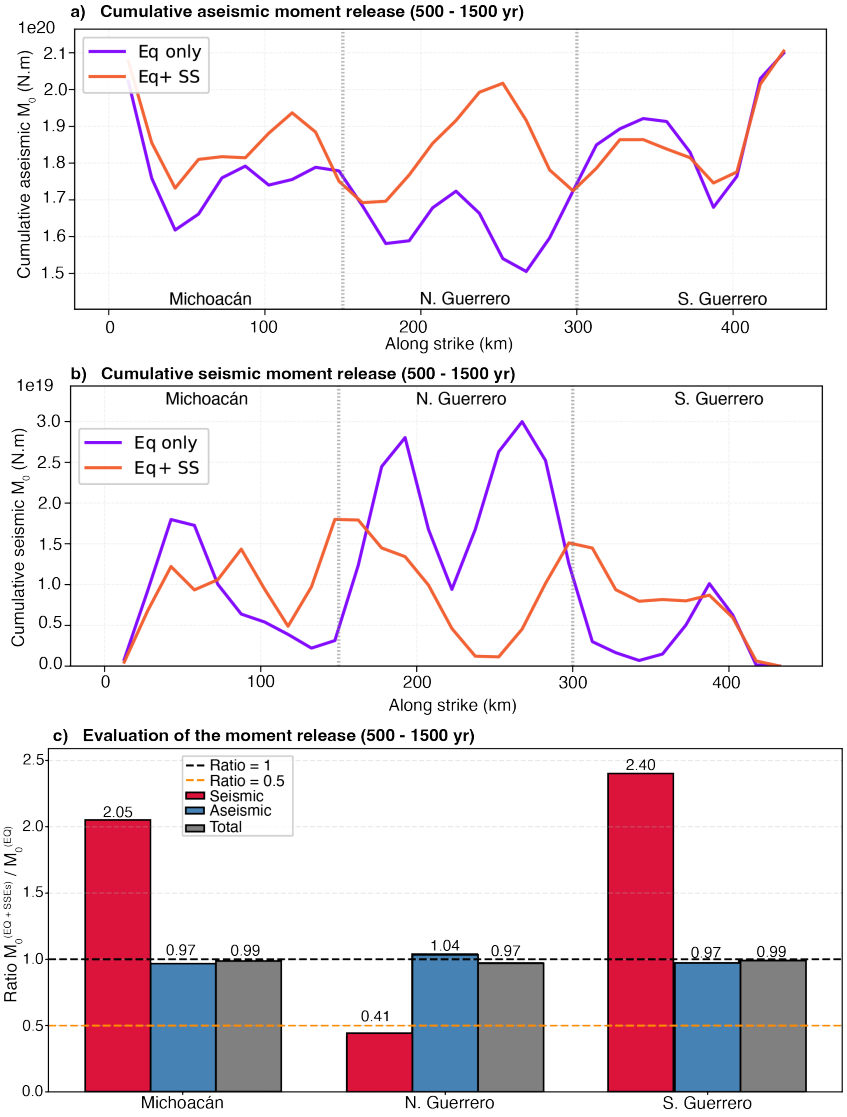


Figure 2. Along-strike distribution of cumulative moment release in the seismogenic zone over 1000 years and many earthquake cycles. (a) Cumulative aseismic moment release ($V < 10^{-3}$ m/s) along strike. The purple line is is for EQ-only simulation and the orange line for the EQ+SS simulation. (b) Cumulative seismic moment release ($V \geq 10^{-3}$ m/s) along strike. Vertical dotted lines and labels mark the three along-strike analysis regions. (c) Ratio of cumulative moment release in the EQ+SS simulation to the EQ-only simulation, computed over the 500–1500 year window (to avoid spin-up effects), for each of the three fault segments. Red bars indicate the moment released during high-velocity (seismic) intervals ($V \geq 10^{-3}$ m/s). Blue bars represent the moment released during low-velocity (aseismic) intervals. Gray bars show the total moment. The dashed horizontal lines show a ratio of 1 (no difference between the two models) and a ratio of 0.5 (proportional to the geodetic coupling reduction derived in Section 4). The total moment is nearly conserved segment by segment (gray bars at 0.97–0.99) as expected from the prescribed long-term loading. The aseismic ratios remain close to one (blue bars at 0.97–1.04), but this does not mean that the aseismic budget is unchanged. Because aseismic slip dominates the total moment release, a few-percent aseismic change is sufficient to balance a much larger relative change in seismic moment. However the seismic moment is strongly redistributed. In northern Guerrero the seismic release is reduced by a factor of 2 (0.41), while neighboring segments release respectively 2.05 and 2.40 times more seismic moment than in the EQ-only case. Robustness of these ratios across time windows is shown in Figure S7. The corresponding two-dimensional spatial redistribution of cumulative seismic moment is shown in Figure S10.

averaging over many earthquake cycles, we can estimate the influence of recurring slow slip on measurements of plate coupling (Text S4).

In the EQ-only simulation, coupling remains uniformly high (0.90–0.95) throughout the interseismic period in all three segments (Figure 3a & c). When slow slip is introduced, coupling in central northern Guerrero is markedly reduced with median values dropping from ∼0.80 to 0.65 (Figure 3e & f). This reduction is associated with recurring slow slip that provokes transient aseismic creep in the neighboring updip seismogenic zone (Figures S11–S13), even as the seismogenic zone remains strongly coupled in between slow slip events. The coupling distribution also broadens. Without slow slip, coupling only varies between 0.85–0.95, whereas the EQ+SS model exhibits values that evolve through time and reach as low as 0.4 (Figure 3f). This variability is also visible in the spatial persistence of high coupling. In the EQ-only model, coupling over the 40–10 yr window exceeds 0.5 before more than 50% of earthquakes across a majority of the seismogenic zone. In the EQ+SS model, this fraction is substantially reduced directly above the slow slip zone in central northern Guerrero (Figure S8).

The seismogenic coupling oscillates with a 4-year periodicity that matches the slow slip cycle. Each slow slip event temporarily reduces coupling on the updip seismogenic segment before coupling gradually recovers (Figure 3b). This is analogous to reported observations of time-varying coupling (Nishimura et al., 2004; Materna et al., 2019; Villafuerte et al., 2025), and implies that plate coupling estimated over modern geophysical time scales of one to several decades only captures a snapshot of the state of the plate interface that evolves with both the slow slip cycle and the broader seismic cycle (Maubant et al., 2023). These oscillations are pronounced in northern Guerrero updip of the slow slip zone, while coupling is more constant through time in the more distant Michoacán and southern Guerrero segments. This localized coupling change does not contradict the enhanced cumulative seismic release in the adjacent segments (Figure 2). Coupling is measured over finite interseismic windows and reflects the slip deficit before earthquakes, whereas the seismic moment budget integrates coseismic slip over many cycles. The adjacent segments release more seismic moment over years 500–1500 because ruptures nucleate, propagate, and arrest differently when slow slip is present. The pre-earthquake coupling in the Michoacán and southern Guerrero segments is nearly identical in the two models, with both segments exhibiting strong coupling ($\phi \sim 0.9$) despite the coupling reduction in the northern Guerrero segment. In such a way, slow slip reorganizes the long-term rupture pattern across the three megathrust segments with a reduction in coupling only manifesting directly updip of the slow slip source.

4.2 Recovered plate coupling from synthetic surface observations

The reduction in coupling is amplified when viewed through the lens of geodetic observations. We generate synthetic GNSS surface displacements from our simulations and invert them for plate coupling with the same approach used for real observations in Guerrero (Radiguet et al., 2016; Maubant et al., 2022, 2023) (Figure 4). We refer to this geodetically-resolved coupling as the recovered coupling, as opposed to the in situ coupling estimated directly on the fault. In the EQ-only model, surface displacement rates are uniform (3.5–4 cm/yr) and the recovered coupling exceeds 0.8 across the seismogenic zone (Figure 4. When slow slip is introduced in the EQ+SS model, displacement rates above northern Guerrero decrease to 2.5-–3 cm/yr, and a distinct zone of poor coupling appears with values reduced by 0.25–0.50 compared to the EQ-only model (Figure 4b & c). This larger reduction relative to the on-fault estimate (0.15–0.20) reflects the limited resolving power of geodetic inversions, and in particular the trade-off between coupling amplitude and fault area. Surface data primarily constrain the moment associated with slip deficit and cannot always distinguish a localized coupling reduction from a broader, lower-amplitude reduction when network coverage is limited. A synthetic time series illustrates this with periodic 4-year oscillations and a systematically lower mean displacement rate in the EQ+SS model com-

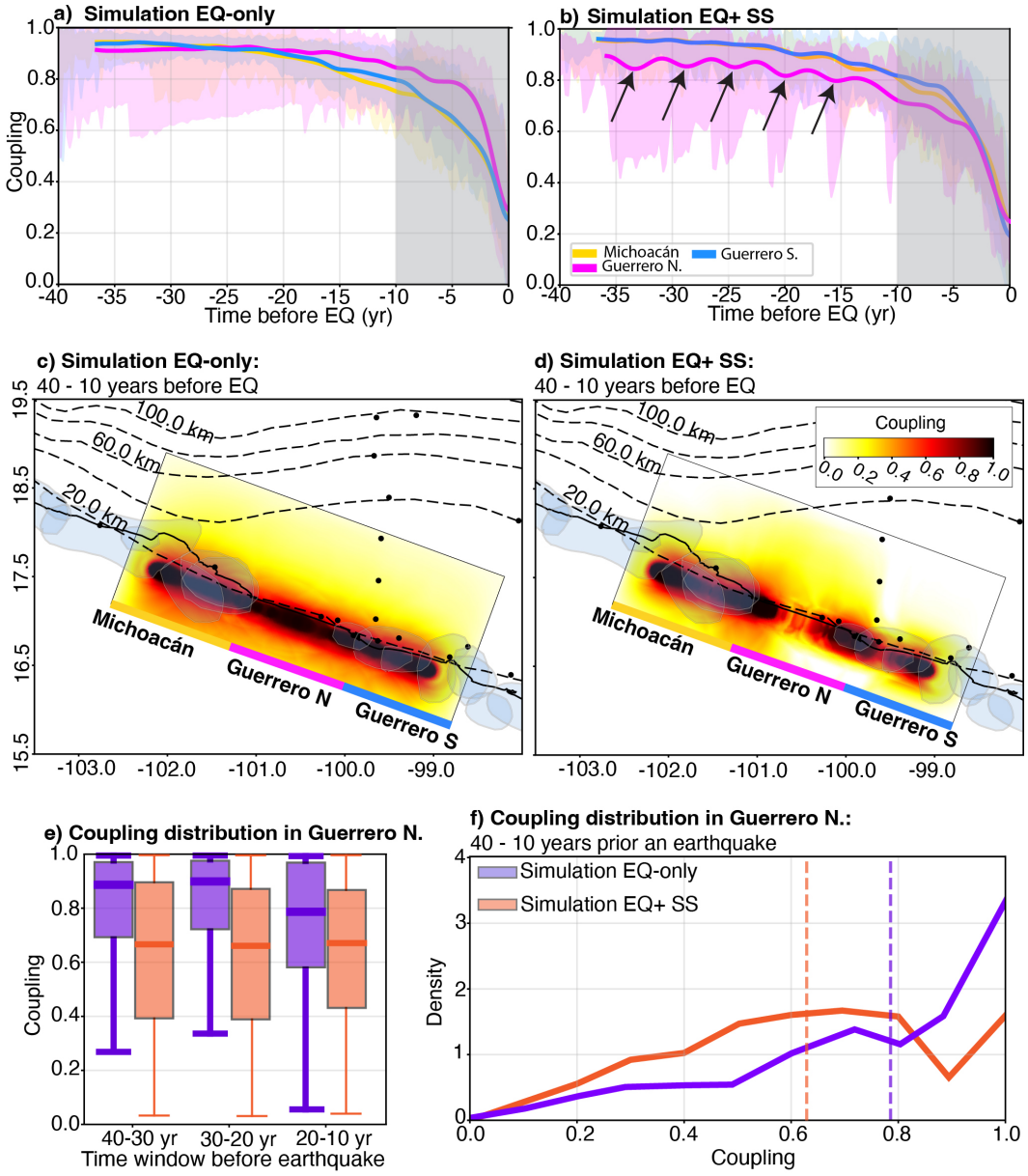


Figure 3. Capturing the influence of slow slip on the spatiotemporal evolution of interseismic coupling prior to earthquakes. (a, b) Coupling time series stacked before each rupture for the three fault segments. The yellow, magenta, and blue lines respectively represent coupling in the Michoacán (0–150 km), northern Guerrero (200–250 km), and southern Guerrero (300–400 km) segments. Solid lines show median values of coupling; the shaded envelopes indicate the associated standard deviations. Gray shading marks the final 10 years before rupture, excluded from the analysis. The EQ-only simulation is shown in (a). The EQ+SS simulation is shown in (b), highlighting the 4-year oscillations in the northern Guerrero segment that correspond to recurring slow slip (indicated by the arrows). (c, d) Median coupling on the fault plane measured 40–10 years before earthquakes for (c) the EQ-only and (d) the EQ+SS simulations. Panels (a–b) and (c–d) show the same quantity, $\phi(t)$ of Eq. 2, respectively averaged in space over each rupture area and averaged in time over the 40–10 yr window. Colored bars along the bottom indicate segment locations. (e) Box-and-whisker plots of coupling values in northern Guerrero for three time windows, comparing EQ-only (purple) and EQ+SS (orange) simulations. (f) Probability density distributions of coupling measured 40—10 years before earthquakes in northern Guerrero; dashed lines indicate the median values: 0.80 for the EQ-only model and 0.65 for the EQ+SS model.

pared to the EQ-only model (Figure 4d & e). Although the slow slip zone is designed to slip aseismically, it exhibits non-zero recovered coupling because earthquake-associated slip occasionally extends into this region (Figure S9), contributing to the measured slip deficit.

Critically, this low recovered coupling does not imply an aseismic fault. We see a fault segment with poor coupling still routinely ruptures in Mw≥7 earthquakes (Figures S5, S9). A geodesist analyzing 20–30 years of GNSS data from this region would infer coupling coefficients of 0.4 to 0.6 in northern Guerrero and a consequently reduced seismic hazard. The relative decrease in seismic moment in northern Guerrero (0.41; Figure 2c) is consistent with the recovered coupling reduction. The weakly coupled segment genuinely releases less seismic moment, but most of the "missing" moment is released by the neighboring segments with the remaining balance as slow slip. Plate coupling estimates recovered from geodetic data thus depend both on the timescale of observations relative to both the slow slip and megathrust cycles (Maubant et al., 2023) and on the position of the segment relative to the slow slip source.

5 Implications for seismic hazard in the presence of slow slip

The Guerrero seismic gap presents a paradox: geodetic observations show low coupling over the last two decades, yet the region has likely hosted large historical earthquakes, including a possible M_w 7.5–8 event in 1911 (Singh et al., 1981; Husker et al., 2023). We find recurring slow slip is associated with transient aseismic creep in the shallower seismogenic zone, reducing the moment ultimately released seismically in the segment directly updip (ratio 0.41, Figure 2c). This localized reduction does not represent a loss of seismic potential at the scale of the megathrust. The deficit is instead largely transferred to neighboring along-strike segments, which rupture more, while the remainder is released as additional slow slip. The segment directly updip of the slow slip zone therefore remains capable of major earthquakes, while neighboring segments may release more seismic moment than expected from a model with only earthquakes. The century since the last historical earthquake that likely happened in 1911 is consistent with slow slip lengthening recurrence intervals in the segment directly updip.

While we focus on Guerrero, the same association between slow slip and low recovered coupling recurs at other margins. Beyond the Hikurangi and Alaska examples noted in the introduction (Wallace & Beavan, 2010; Bell et al., 2014; Maubant et al., 2023; He et al., 2023), a compilation across Nankai, Alaska, Hikurangi, and Mexico finds regularly recurring slow slip adjacent to or within weakly coupled regions (Takagi et al., 2019), and deep slow slip with a ∼5-year return time underlies the intermediate-coupling Atacama region of Chile (Klein et al., 2018, 2023). This relationship is not universal, however: in Cascadia (Dragert et al., 2001) and in the Oaxaca segment of Mexico (Rousset et al., 2016), slow slip occurs while geodetic coupling remains high. Why some regions with slow slip exhibit low coupling updip while others do not remains an open question. Potential factors that control this relationship include the spatial relationship between slow slip source regions and the seismogenic zone, the rheology of the plate interface, and the resolution of geodetic inversions (Y. Liu & Rice, 2005; Rousset, 2019; Takagi et al., 2019). In our model the slow slip source is adjacent to the seismogenic zone, and we find the coupling reduction is strongest immediately updip of slow slip. This suggests that the separation between the source and the seismogenic zone may be an important control, although its quantitative influence remains to be established through further sensitivity tests. Where station spacing is sparse and smoothing length-scales exceed the dimension of coupled fault segments, coupling can be underestimated independently of the mechanical interplay documented here. Our results therefore do not imply that all seismogenic zones updip of slow slip will appear weakly coupled, but rather that when low coupling is observed in the presence of slow slip, it should not be read as evidence of a quiescent fault.

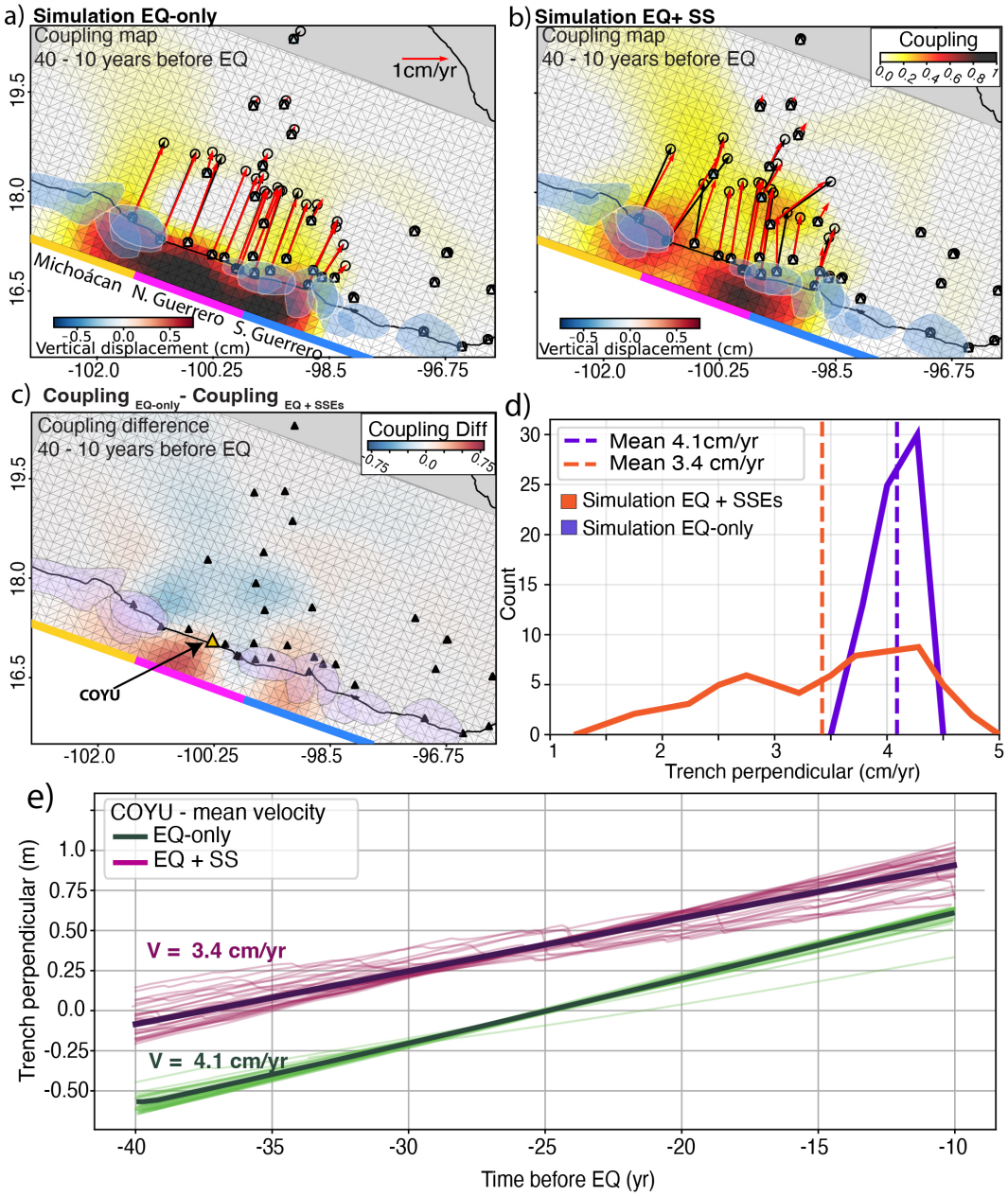


Figure 4. Recovered maps of plate coupling obtained by inverting synthetic surface displacements. (a) Inversion of the synthetic data from the EQ-only simulations to recover the coupling plate. (b) same as (a) but for EQ+SS simulation. (c) Maps of coupling differences (EQ-only coupling minus EQ+SS coupling). Positive values (red) indicate regions where slow slip events reduce recovered coupling. Red arrows show forward-modeled horizontal displacements from the coupling inversion. Black arrows show the synthetic observations used as input. Triangles and squares indicate vertical displacements from the forward model and synthetic observations, respectively. The historical ruptures are represented in cyan (Kostoglodov & Pacheco, 1999). (d) Histograms of the velocity estimates from the synthetic time series (COYU) in the trench-perpendicular direction computed 40–10 years before earthquakes. The purple line represents the EQ-only simulation (mean: 0.041 m/yr) and the orange line the EQ+SS simulation (mean: 0.034 m/yr). We find slow slip reduces the average surface velocities by 15-17% and make the interseismic period more variable. (e) Surface displacement time series, trench perpendicular, at station COYU (yellow triangle in panel c) stacked relative to earthquake occurrence. Individual cycles are shown as thin lines, with green representing the EQ-only model and magenta the EQ+SS simulation. The bold lines indicate the mean velocity.

Several simplifications warrant caution in extrapolating these results to regional hazard assessments (Text S6). First, our model neglects stress interactions with regions outside the modeled fault geometry (Zigone et al., 2012; Cruz-Atienza et al., 2021). Also the model geometry represents a simplified megathrust with several piecewise-uniform frictional domains (Cruz-Atienza, Tago, et al., 2025) and does not include the shallow slow slip recently documented offshore Guerrero (Plata-Martínez et al., 2021; Cruz-Atienza, Tago, et al., 2025) that may load the seismogenic zone from updip. Despite a simplified geometry, our results show that geodetic coupling inferred from decadal surface velocities cannot be interpreted as a direct proxy for seismic potential in subduction margins with recurring slow slip. Low recovered coupling can thus reflect a reorganized moment budget rather than a quiescent fault, motivating hazard assessments at the scale of the fault system rather than individual segments.

6 Open Research

The seismic cycle simulations in this study were performed with the Unicycle software (Moore et al., 2019; Barbot, 2019), available at `https://bitbucket.org/sbarbot/unicycle/src/master/` and archived on Zenodo (`https://doi.org/10.5281/zenodo.5688288`. The results of the simulations can be found here: `https://zenodo.org/uploads/23179607`).

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